

# DISCO2VERY – Ethylenerouten

25.06.2026

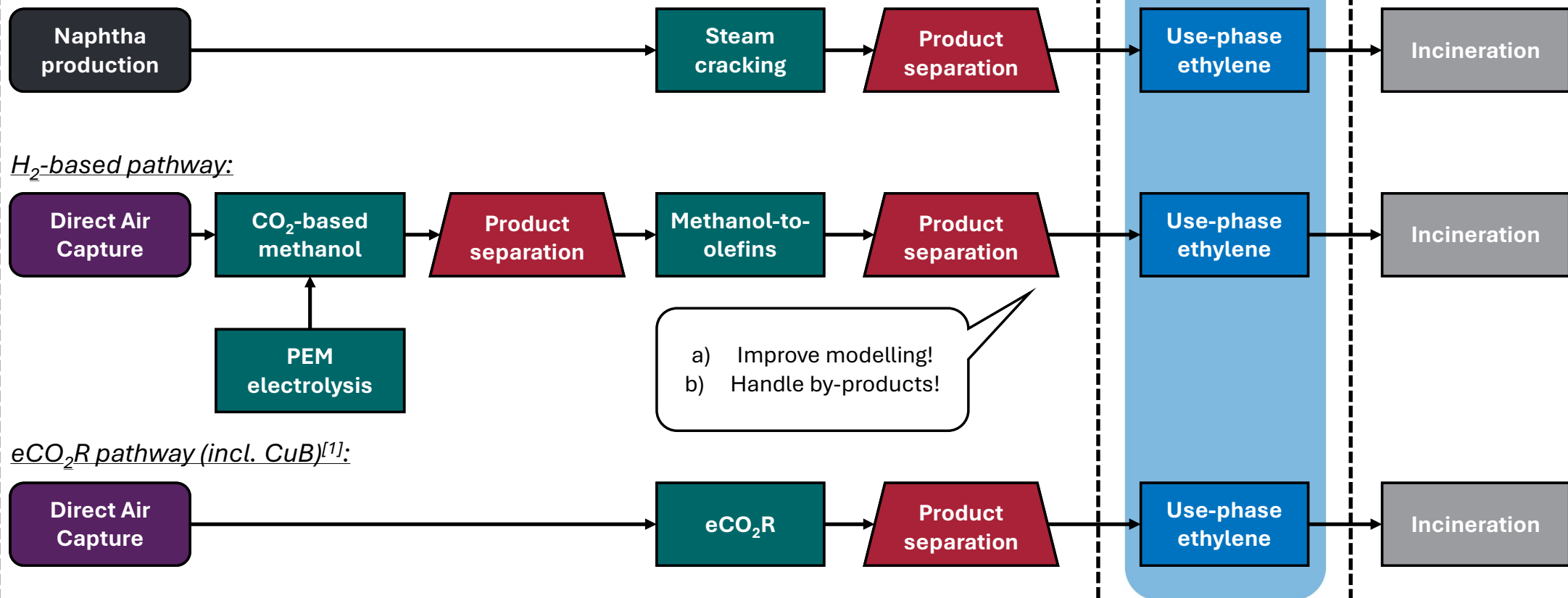
Alexis Spohr

## Approach of LTT

### Scope of the study: pathways for ethylene production



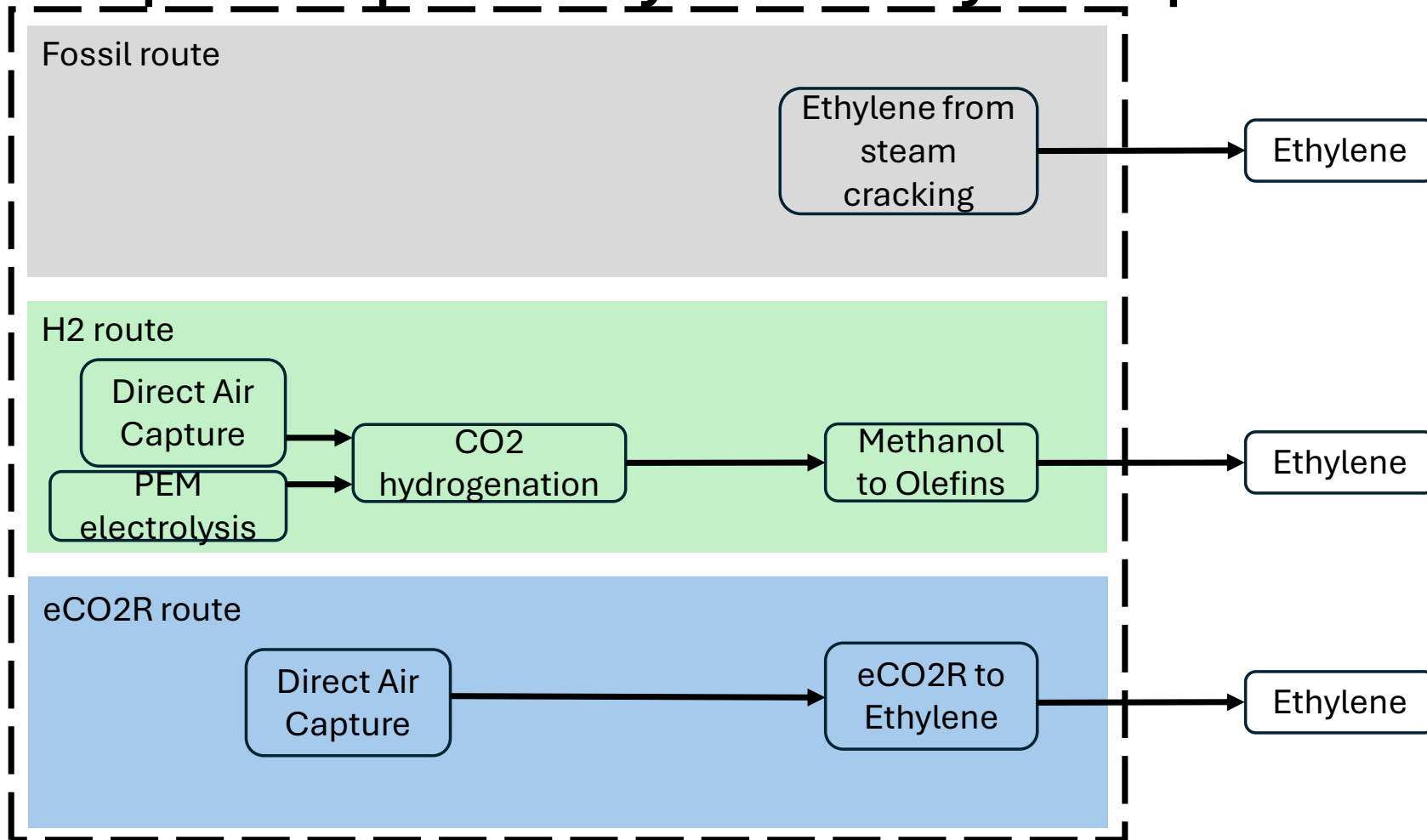
#### Fossil-fuel-based pathway:



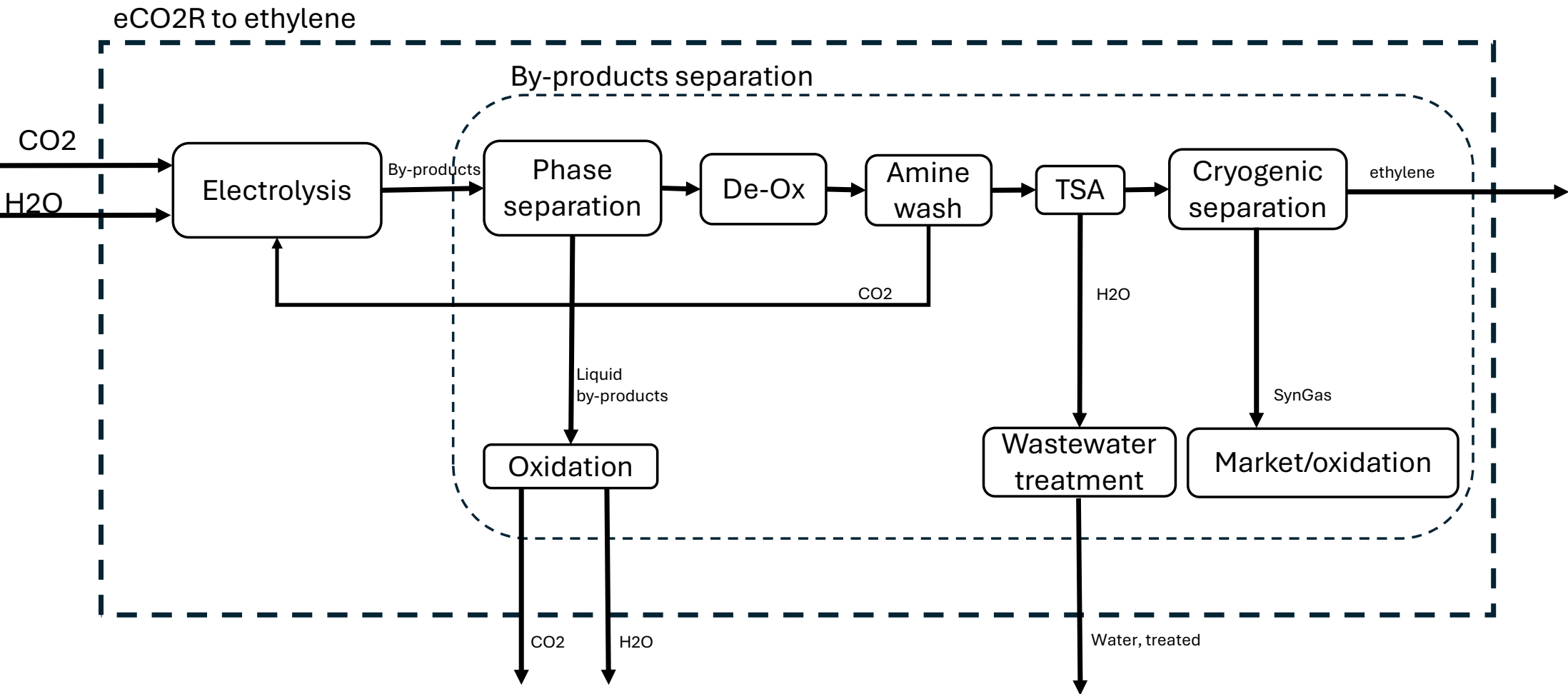
System boundary

[1] Löffelholz et al. Optimized scalable CuB catalyst with promising carbon footprint for the electrochemical CO<sub>2</sub> reduction to ethylene. 2023.

# Simplified pathways for ethylene production



# eCO<sub>2</sub>R to ethylene in detail

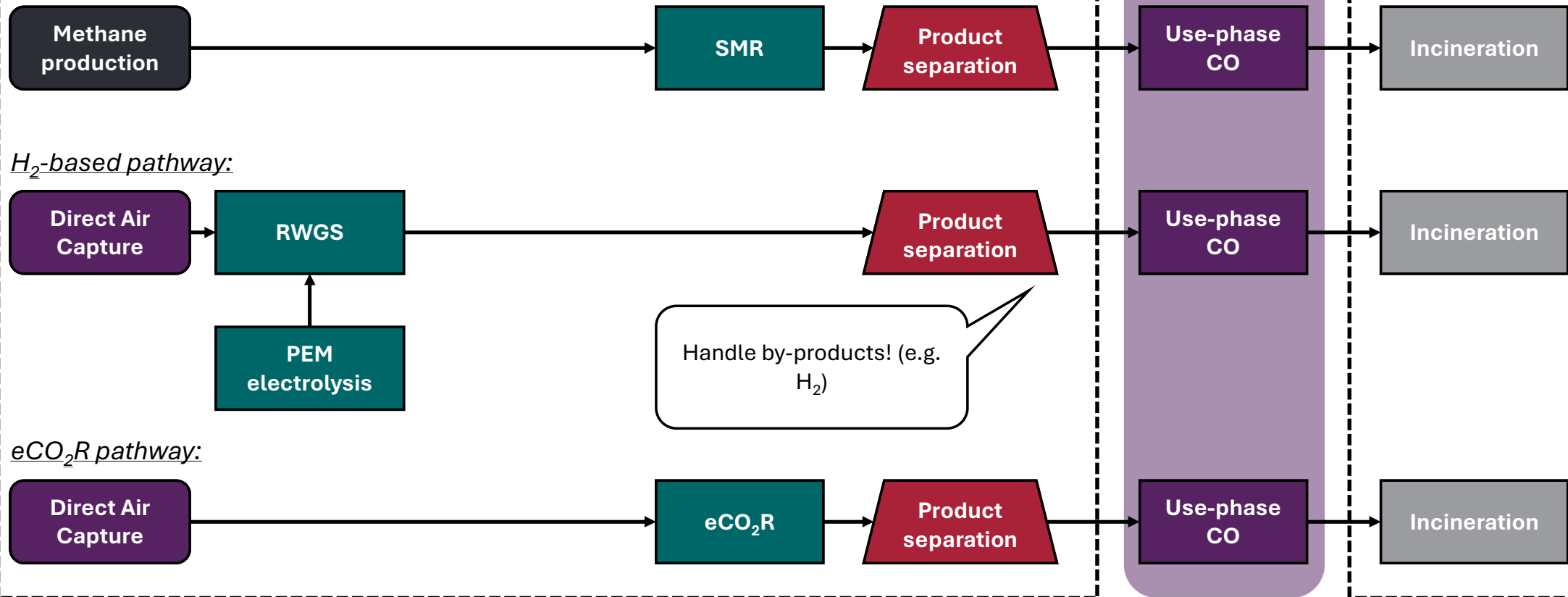


## Approach of LTT

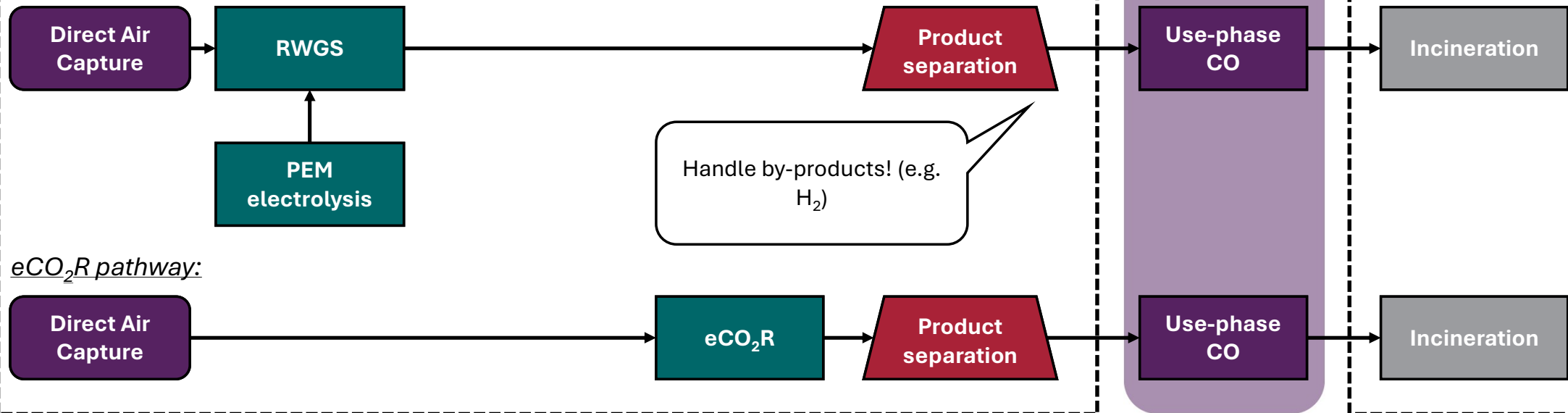
### Scope of the study: pathways for ethylene production



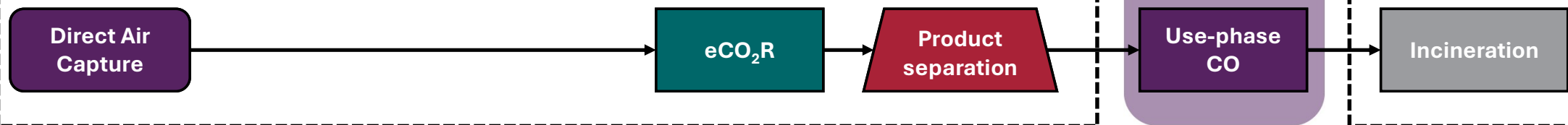
#### Fossil-fuel-based pathway:



#### H<sub>2</sub>-based pathway:



#### eCO<sub>2</sub>R pathway:



System boundary

RWGS: reverse water-gas-shift  
SMR: steam methane reforming